

Learning Day 9 - Project Flow Infographic

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Learning Day 9 — Ensemble Methods Project Flow

What we completed from Notebook 1 to Notebook 6

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Notebook 1 — Source Audit & Plan

- Audited the uploaded course files
- Built a coverage matrix for all required topics
- Selected the Bank Marketing dataset
- Defined the Day 9 notebook roadmap and evaluation plan

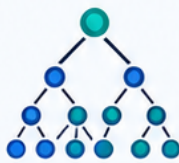
2



Notebook 2 — Data Preparation & Baselines

- Explored target balance, missing values, and feature types
- Created train/test split and preprocessing pipeline
- Trained baseline models: Dummy, Logistic Regression, Decision Tree, KNN
- Saved reusable data and baseline outputs

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Notebook 3 — Bagging & Random Forest

- Compared Single Decision Tree vs Random Forest (100 trees)
- Reviewed feature importance for both models
- Tested n_estimators from 10 to 500 and found stagnation around 100
- Checked OOB score and Random Subspace Method via max_features
- Confirmed Random Forest clearly outperformed a single tree

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Notebook 4 — Cross-Validation & StackingClassifier

- Ran 5-fold cross-validation for the base learners
- Built a StackingClassifier using RF, Logistic Regression, KNN, and Decision Tree
- Compared Logistic Regression vs Random Forest meta-models
- Studied model diversity and compared CV + held-out test results
- Best held-out stacking model: RF meta-model

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Notebook 5 — Manual OOF Stacking & Blending

- Built manual 5-fold Out-of-Fold prediction features
- Trained a meta-model on OOF meta-features
- Built Blending with a 70/30 split
- Compared OOF Stacking vs Blending in performance and stability
- Found Manual OOF Stacking stronger than Blending

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Notebook 6 — Three-Layer Stacking & Gradient Boosting

- Created a 3-layer stacking system: Level 0, Level 1, Level 2
- Evaluated the final Three-Layer Stacking model
- Added a compact Gradient Boosting model for theory-to-practice coverage
- Compared complexity vs business value
- Practical recommendation: simpler RF-meta stacking remained the best choice



Outcome So Far

- ✓ All major Day 9 ensemble topics from Notebook 1 to 6 were covered
- ✓ Practical methods included Bagging, Random Forest, OOF Stacking, Blending, Three-Layer Stacking, and Gradient Boosting
- ✓ Best practical model so far: StackingClassifier with Random Forest meta-model